

# Code-quality and frontend QA report

Repository: kortix-ai/suna

Reviewed: 2026-08-28

Original head: 5c609e689e03f2a93468e32257ac537e7e1f1517

Fix commit: 7cc805c37578d66c50ef9ebed1c76c58ed45eb64

Reviewed code head: 1f86186e6d0c9c526960e9b18834e65395e7d1fc

## DECISION: DO NOT MERGE YET

The confirmed code gaps are fixed. Queue persistence and refresh visibility pass in Chromium. The cloud runtime ended with runtime\_gone before the second prompt completed. New-head CI is still running.

## Executive result

- JAY-720 now retries history from the last successful cursor.
- JAY-721 now bounds the first never-settling /start request.
- The lifecycle queue now sends one prompt per session.
- Empty zero-token assistant rows no longer paint blank turns.
- Older empty queue snapshots cannot erase a newer queue row in the SDK observation path.
- The branch is synchronized with main. GitHub reports mergeable: true.

## Confidence

Code-level confidence is high for the fixed regressions. User-flow confidence is medium. Ship confidence remains blocked by one stable two-prompt completion and new-head CI.

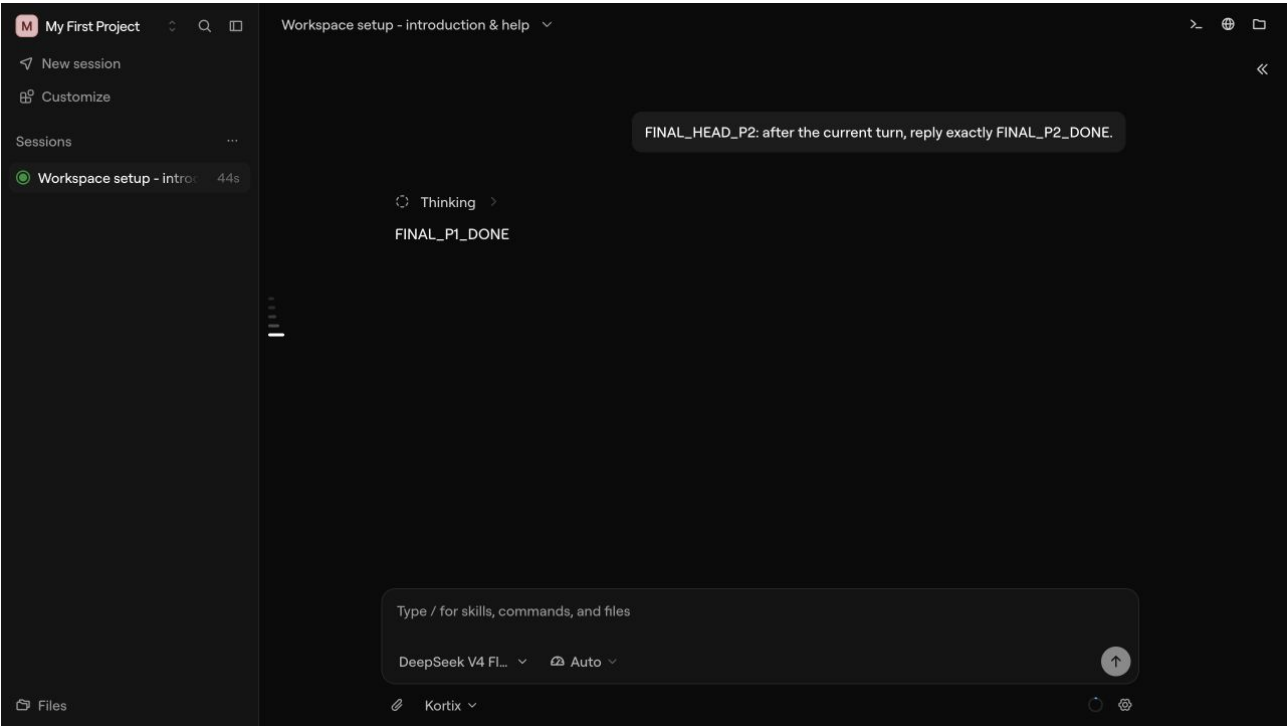
# Original-head browser failure

P1 requested a 30-second wait. P2 was submitted while P1 was active. Both lifecycle rows were marked delivered, but the first answer appeared below P2. The requested P2 result never appeared.

| Prompt | Started UTC  | Ended UTC    | Duration |
|--------|--------------|--------------|----------|
| P1     | 17:43:26.534 | 17:43:59.945 | 33.411 s |
| P2     | 17:43:38.791 | 17:44:01.254 | 22.463 s |

## Confirmed integrity failure

The turns overlapped for about 21.154 seconds. The UI attributed FINAL\_P1\_DONE to P2. No assistant message contained FINAL\_P2\_DONE.



Original PR head: the P1 result appears below the P2 prompt.

# Fixes implemented

## JAY-720 - history cursor

- Each successful page hydrates messages and commits its cursor.
- A failure after cursor 5 now retries with before=cursor-5.

## JAY-721 - first unresolved start

- useSessionStartGiveUp owns the deadline and verdict.
- The first unresolved request arms the deadline.
- Later data or errors clear the deadline and verdict.

## Durable queue serialization

- An active turn refuses admission with reason turn\_active.
- One drain sends only the head prompt for each session.
- Sibling prompts return to the queue with bounded backoff.
- The engine no longer starts same-session deliveries concurrently.

## Related correctness fixes

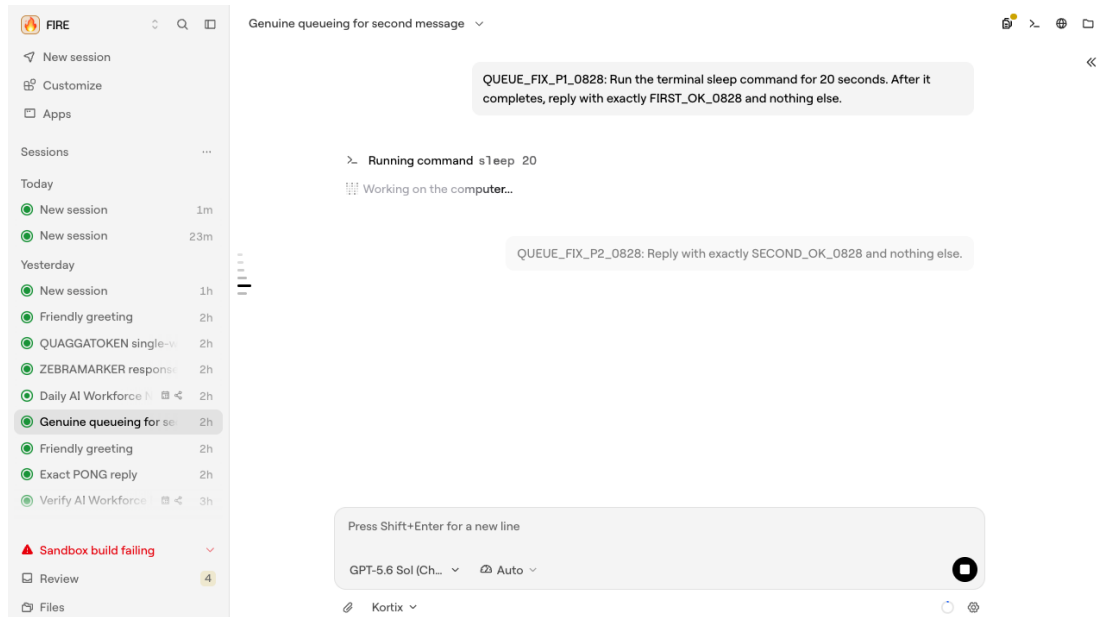
- Parentless zero-part, zero-token assistants are skipped only when they have no error.
- Queue projection and query cache updates share applyInboxObservation.
- splitTimingDurations removes wall-clock subtraction from timing assertions.

## Post-main-sync regression gates

Full SDK suite: pass. SDK typecheck: exit 0. Focused API queue and timing tests: 38 pass, 0 fail.

## Post-fix browser evidence

P1 requested a 20-second wait and FIRST\_OK\_0828. P2 requested SECOND\_OK\_0828 while P1 was active. Both POST /prompts calls returned HTTP 202. GET /prompts retained P2 with reason turn\_active and attempts 0.



*P2 remains queued while P1 owns the active turn.*



The queued P2 row remains visible after a browser refresh.

# Browser verdict

## Verified

P2 stayed in the durable queue. The UI displayed its queued state. A refresh preserved the row. No same-session overlap occurred during the observed window.

## Not verified

The runtime ended with `runtime_gone`. P1 was requeued as redelivered after `runtime_gone`. P2 remained waiting. The run did not prove automatic P2 completion or final parent-to-answer attribution.

# Automated verification

| Surface                         | Result                      |
|---------------------------------|-----------------------------|
| Focused SDK regression files    | 71 pass, 0 fail             |
| Mounted start give-up hook      | 2 pass, 0 fail              |
| Full SDK suite after main sync  | Pass                        |
| SDK typecheck after main sync   | Exit 0                      |
| SDK packed install smoke        | Exit 0                      |
| Focused API after main sync     | 38 pass, 0 fail             |
| Full API before main sync       | 8,748 pass, 79 skip, 0 fail |
| Full web before main sync       | 8,648 pass, 0 fail          |
| Standalone CLI before main sync | 1,243 pass, 0 fail          |
| Changed-file ESLint             | 0 errors, 32 warnings       |
| git diff --check                | Clean                       |

The root packages-only run failed in its concurrent CLI sub-lane without useful output. The same CLI suite passed standalone. Treat the root package lane as unresolved until CI completes.

# Linear Phase 0 verification

| Issue   | Assessment                                             |
|---------|--------------------------------------------------------|
| JAY-717 | Unit-covered; real provider retry not induced          |
| JAY-718 | Long-turn Stop verified; provider retry not induced    |
| JAY-719 | Keepalive excluded from process evidence               |
| JAY-720 | Fixed and verified                                     |
| JAY-721 | Fixed and verified                                     |
| JAY-722 | Active provisioning browser verified                   |
| JAY-723 | Unit-covered; real poison sequence not induced         |
| JAY-724 | Boot-phase header browser verified                     |
| JAY-725 | Cache-key covered; custom directory browser run absent |
| JAY-726 | Empty assistant suppression fixed by tests             |
| JAY-291 | Legacy queue serialized; protocol v2 remains           |

## Status discipline

All PR-related issues remain In Progress. Repository policy permits Done only after merge, dev deployment, and dev verification.

## New tracked gaps

### JAY-728 - queue snapshot freshness clock

POST, GET, bundle, and stream observations need one server-defined order or revision.

### JAY-729 - failed-session redirect escape

A failed session needs a deterministic browser test, a visible escape action, and no redirect loop.

## Marko critical-issue mapping

| Items | Theme                            | Tracking / assessment                 |
|-------|----------------------------------|---------------------------------------|
| 1-3   | Mid-turn death and silent death  | JAY-430, JAY-566, JAY-723; not closed |
| 4     | 140 MB history and OOM           | JAY-686, JAY-689, JAY-351; not closed |
| 5     | UI slower than terminal          | JAY-697; not closed                   |
| 6     | Agent and stream architecture    | Protocol phases 1-4; Phase 0 only     |
| 7-10  | Queue and reconnection           | JAY-291, JAY-572, JAY-726; improved   |
| 11    | Warm prior-session regression    | JAY-596; not closed                   |
| 12    | Slow resume                      | JAY-697; not closed                   |
| 13    | Redirect softlock                | JAY-729; new urgent ticket            |
| 14-15 | Stop/restart and bricked session | JAY-430, JAY-566, JAY-723; not closed |
| 16    | Unexplained terminal incident    | Needs incident-specific evidence      |

### Scope conclusion

PR #6957 is Phase 0 symptom relief. It does not complete the Session State Protocol architecture or close all 16 incidents.

# Thermo-nuclear code-quality review

## Correct improvements

- The new start give-up hook removes timing state from the 1,500-line use-session hook.
- Queue observations pass through one helper per observation path.
- The false stream-attached polling mode is removed.
- The terminal card gate uses the SDK SessionStartStage type.
- Reconciler comments now match the actual four subsystem reads.

## Remaining structural debt

| Production file           | Lines |
|---------------------------|-------|
| apps/web session-chat.tsx | 5,433 |
| daemon main.ts            | 3,234 |
| SDK sync-store.ts         | 2,723 |
| lifecycle engine.ts       | 2,244 |
| SDK use-session.ts        | 1,521 |
| session route page.tsx    | 1,417 |

### Review boundaries

No PR file crossed from below 1,000 lines to above 1,000 lines. Source-text assertions remain weaker than mounted or black-box behavior tests. The unrelated globals.css text-blur removal should move to a separate PR.



# Final shipping gate

Do not merge PR #6957 until every item below passes.

- GitHub checks pass for the final PR head after the report commit.
- A stable real sandbox completes P1 and P2 automatically.
- The DOM contains both exact assistant outputs under the correct prompts.
- Lifecycle timestamps prove no same-session overlap.
- Refresh during P1 preserves P2 and later drains it.
- JAY-728 and JAY-729 remain tracked until their acceptance tests pass.
- After merge, dev deploys the merged SHA and the same browser journey passes on dev.

## Current recommendation

Keep the PR open and unmerged. The confirmed code defects are fixed. Wait for CI and rerun the two-prompt browser journey on a stable runtime.

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No credentials are stored in this PDF. No PR merge occurred.